Biochemistry Cume September 15, 2000

Consider the following mixture of peptides:

1. Gly-Ala-Pro-Val-Phe
2. Gly-Asp-Pro-Glu-Phe
3. Gly-Cys-Pro-Cys-Phe
4. Gly-Asn-Pro-Glu-Phe
5. Gly-Asn-Pro-Gln-Phe
6. acetyl-Gly-Asp-Pro-Glu-Phe

7. Gly-Cys-Pro-Cys-Phe
8. Gly-Ala-Pro-Val-Phe-amide
9. acetyl-Gly-Ala-Pro-Val-Phe
10. Gly-Ala-Lys-Val-Phe-amide
11. Gly-Ala-Lys-Val-Phe
12. Gly-Ala-Arg-Val-Phe-amide

Questions:

1. (40 points) The above mixture of peptides is separated by isoelectric focusing. Predict the order of the peptides at the end of the separation. (Some ties may be possible.) Begin with the peptide having the lowest predicted isoelectric point (pI).

2. (10 points) Draw a complete chemical structure for peptide 11 in its predominant ionization state at pH 7. Do not worry about the stereochemistry here (but see #5, below).

3. (20 points) Name or draw a reagent that will convert peptide 7 to peptide 3. Show one or more chemical reactions that illustrate the mechanism of this conversion.

4. (15 points) When stored in a physiological buffer solution at room temperature, peptide 4 is found to decompose slowly. Analysis later shows that the main decomposition product is one of the other peptides in the mixture. Name this peptide, and explain the decomposition reaction.

5. (15 points) Of the 20 common genetically encoded amino acids, 17 of them have exactly one chiral center. Name the 3 exceptions and state why each of them is stereochemically exceptional.